AI & Prescriptive Analytics in Business

BANK TELEMARKETING

Predicting Term Deposit Subscriptions

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PORTUGUESE BANK DATASET

41,188 Clients | **21** Features | Analysis

Agenda

WHAT WE'LL COVER

01
The Dataset
41,188 phone call records from a Portuguese bank.

02
Class Imbalance
Why 88.7% 'No' makes this problem tricky to solve.

03
Who Says Yes?
Job type, education, and marital status patterns.

04
When to Call?
Monthly subscription rate and seasonal patterns.

05
Call Duration
Does longer always mean better? Analyzing talk time.

06
Economic Signals
How market conditions affect customer's decisions.

07
Key Findings
Top predictors and model performance results.

08
Recommendations
Actionable steps for the bank to improve conversion.

Overview

THE DATASET

41,188
Client Records

21
Total Features

20
Input Variables

Numerical Features (10)

- age
- duration
- campaign
- pdays
- previous
- emp.var.rate
- cons.price.idx
- cons.conf.idx
- euribor3m
- nr.employed

Categorical Features (10)

- job
- marital
- education
- default
- housing
- loan
- contact
- month
- day_of_week
- poutcome

Target Variable (y)

Did the client subscribe to a term deposit?

NO | YES

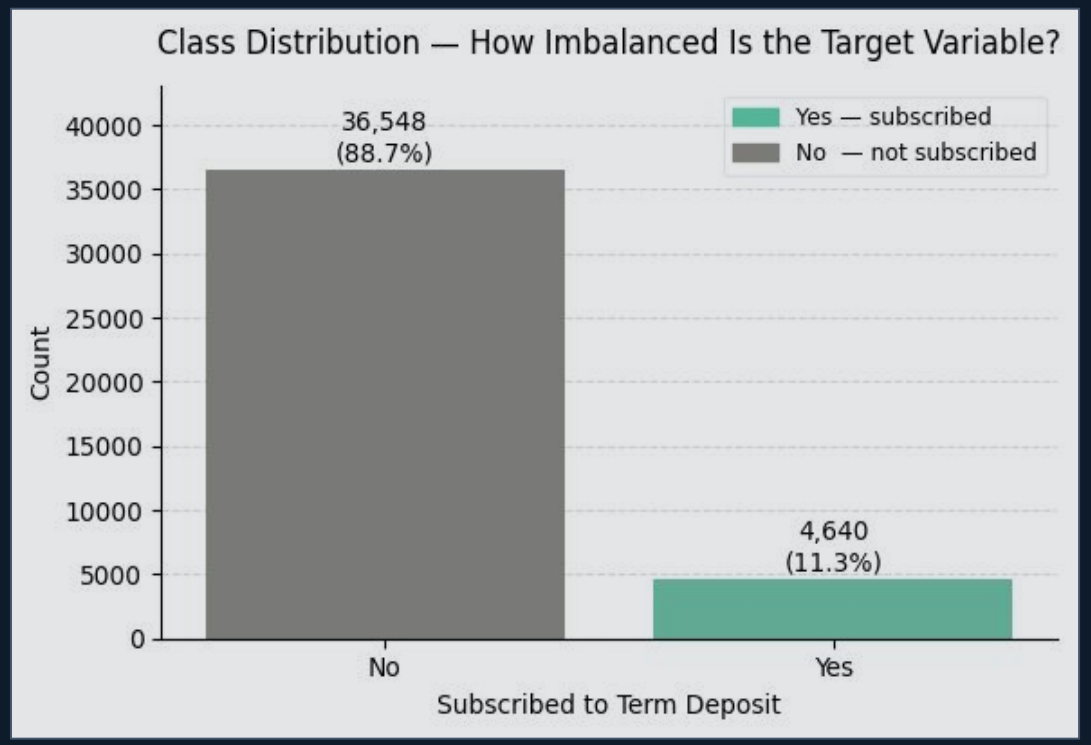
Data Challenge

CLASS IMBALANCE

NO (36,548) YES (4,640)

88.7%
Majority Class (No)

11.3%
Minority Class (Yes)



The Accuracy Trap

Raw accuracy is misleading here. A naive model that predicts "No" for every single client would still achieve **88.7% accuracy**, yet fail completely at identifying potential subscribers.

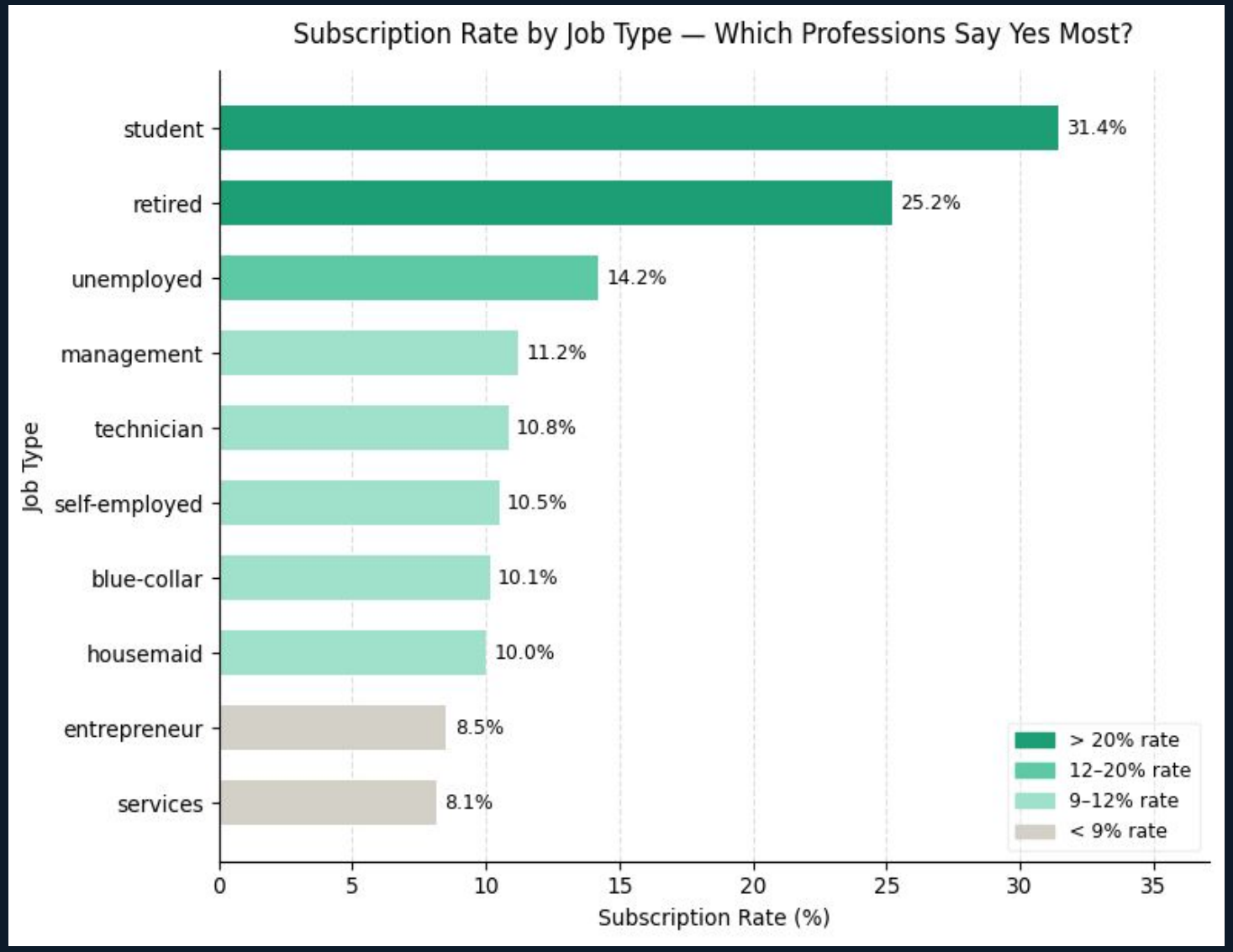
Solution: SMOTE

Synthetic Minority Oversampling Technique was applied to balance the training data.

```
from imblearn.over_sampling import SMOTE
```

Demographics Analysis

WHO SAYS YES?



Key Insight

3.9x Higher Probability

Students and **Retired** clients are significantly more likely to subscribe compared to the blue-collar baseline.

Top Performing Segments

Students	31.4%
Retired	25.2%

Multivariate Context

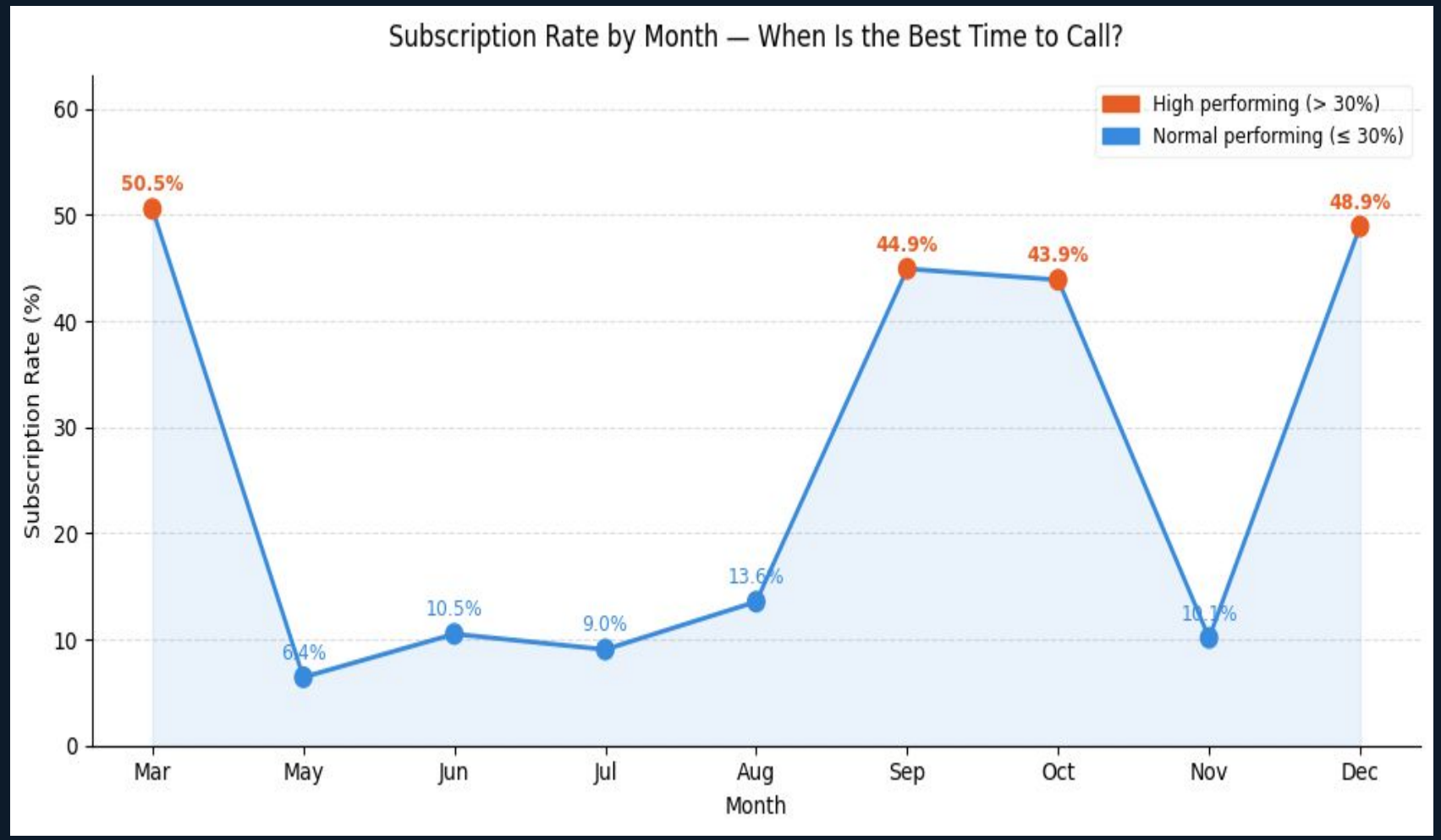
While job type is a strong predictor, **Education** and **Marital Status** also show strong correlation with conversion rates.

Data based on 41,188 client records

y-axis: Job Categories | x-axis: Subscription Rate (%)

Temporal Analysis

WHEN TO CALL?



Best Months

High conversion rates observed in late Q3, Q4, and March.

- Mar 50.5%
- Oct 43.9%
- Sep 44.9%
- Dec 48.9%

Strategic Error

May 6.4%

Lowest conversion rate, yet historically the month with the highest call volume.

ROI Strategy

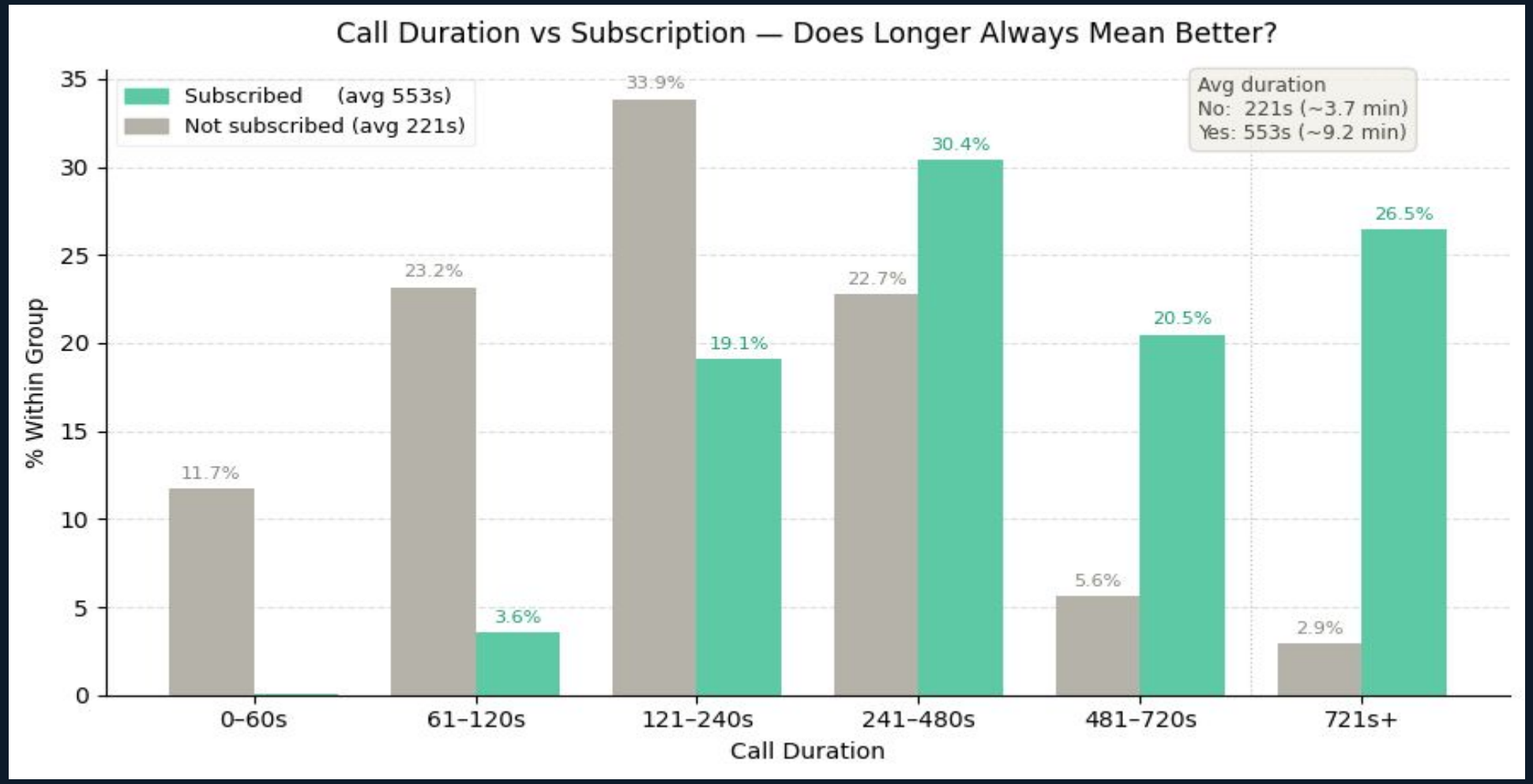
Shift campaign resources to **Q4 (Oct-Dec)** and **March** to maximize term deposit subscriptions.

Vertical bars represent subscription success rate (%) per month

Highlight: Best Months vs Worst Month

Feature Analysis

CALL DURATION vs SUBSCRIPTION



Note: Longer calls indicate genuine engagement and interest. However, duration is a post-call feature you only know it after the call ends, making it a leaky predictor for real-time decision making.

Feature Analysis

CALL DURATION MATTERS

Says NO

Unsuccessful Outcome

221s

~3.7 minutes average

2.5x
Longer

Says YES

Successful Subscription

553s

~9.2 minutes average

Note: Duration is a post-call predictor. While highly correlated, it is not known before the call is made.

Behavioral Analysis

PAST BEHAVIOR PREDICTS FUTURE ACTION

HIGH VALUE

65.1%

Previous Subscribers

Clients who subscribed to a term deposit in a prior campaign.

Higher Conversion
↑ 6.9x

BASELINE

9.4%

New Contacts

Clients being contacted for the first time or with no prior success.

Highest ROI Strategy

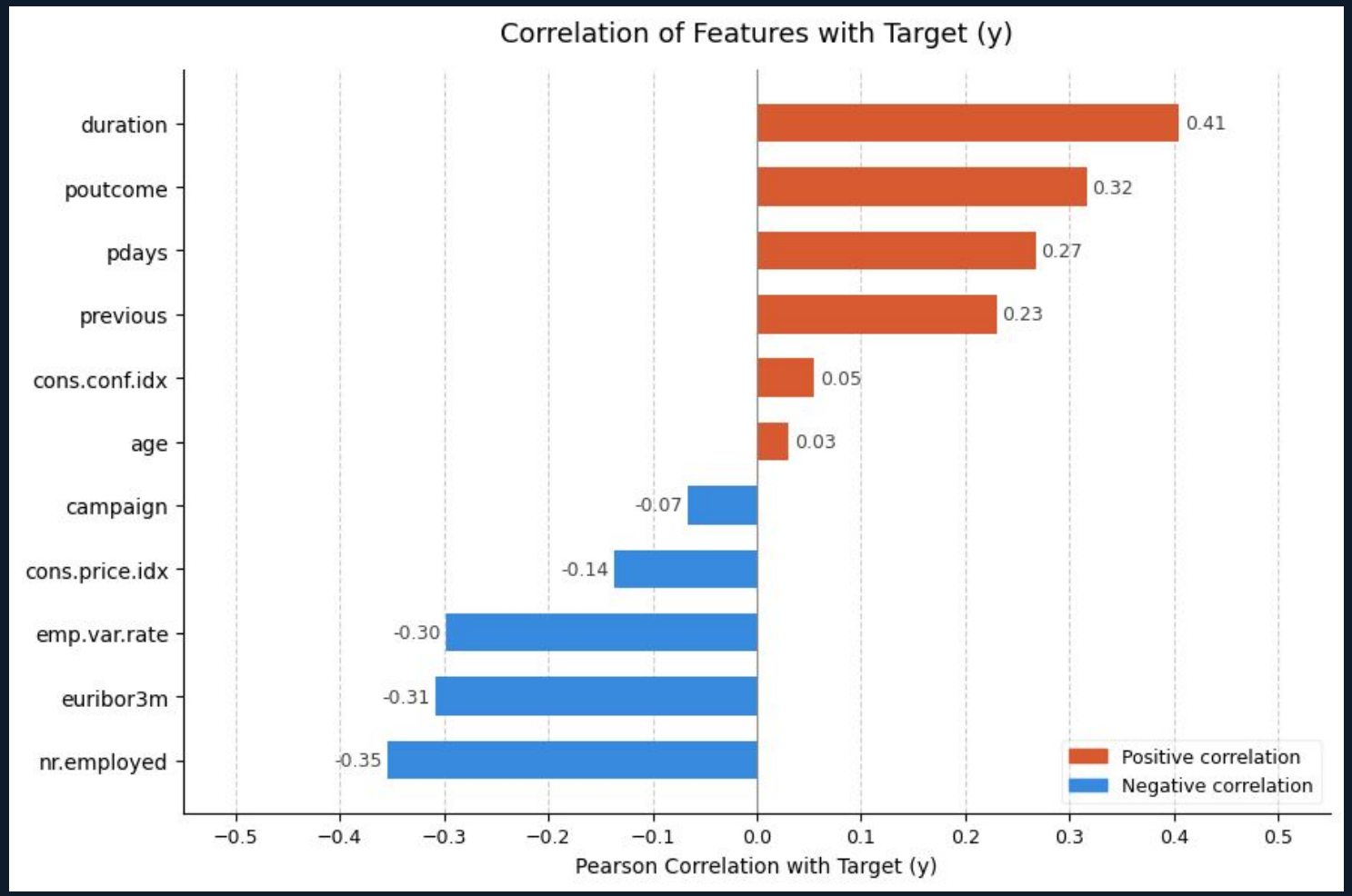
Re-engaging past subscribers is the single most effective tactic. While the volume of these clients is lower, their conversion probability is nearly **7 times higher** than cold outreach.

Predictive Power

The ***'poutcome'*** (previous outcome) variable has a correlation coefficient of **r = 0.32** with the target, making it the second strongest predictor after duration.

Macro Analysis

ECONOMIC SIGNALS



Market Conditions

Economic stability paradoxically reduces term deposit interest. When employment rates and interest rates are **HIGH**, clients are **LESS** likely to subscribe.

↓ `nr.employed (r = -0.35)`

↓ `euribor3m (r = -0.31)`

The Outlier

Unlike macro factors, **Call Duration** is the strongest positive predictor. It represents engagement rather than economic climate.

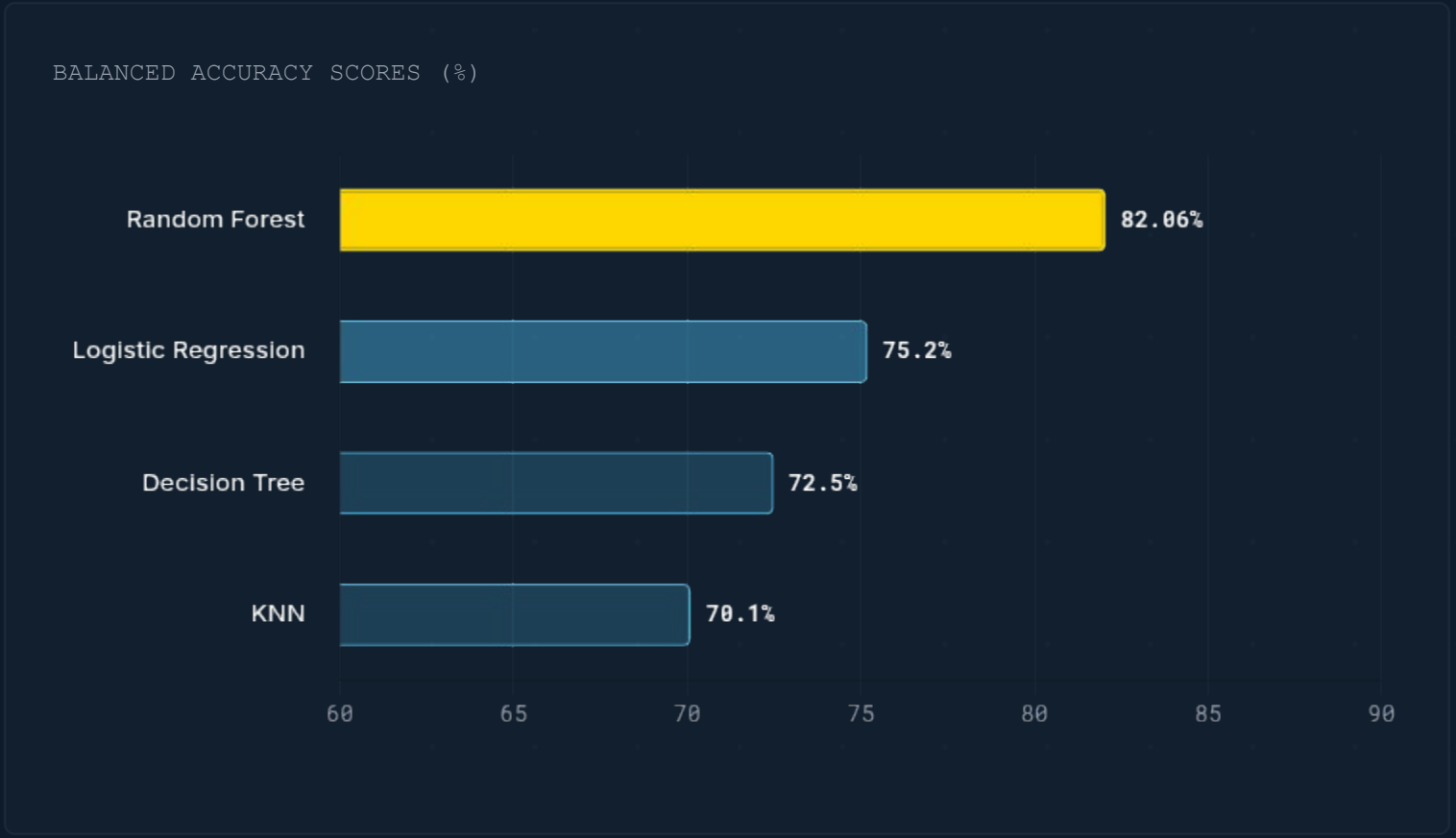
↑ `duration (r = +0.41)`

Pearson Correlation Coefficient (r)

Positive (Red) vs Negative (Blue) Correlation

Model Evaluation

MODEL PERFORMANCE



Pre-processing Technique
SMOTE (Synthetic Minority Oversampling) + PCA applied to handle imbalance.

Validated

CHAMPION MODEL

Random Forest

Classifier (n_estimators=100)

Balanced Accuracy

82.06%

- Best handling of non-linear data
- Robust against overfitting
- Effective with categorical features

Predictive Insights

KEY FINDINGS

01 Call Duration

r = +0.41

The strongest predictor of success. Calls exceeding 5 minutes have significantly higher conversion rates, indicating customer interest.

02 Previous Outcome

r = +0.32

Past success predicts future behavior. The 'poutcome' variable is critical—clients who subscribed before are likely to do so again.

03 Seasonality

High Variance

Timing is key. March, September, October, and December show distinct spikes in success rates compared to the summer slump.

04 Job Type

Demographic

Students and retired individuals outperform other segments. This suggests targeted campaigns for these groups yield better ROI.

05 Economic Indicators

Macro

Employment rates (nr.employed) and Euribor rates negatively correlate with subscriptions. Market stability paradoxically lowers intent.

Fatigue Warning

Excessive contact attempts correlate negatively with success. More calls do not equal more sales; they often lead to customer fatigue and rejection.

r = -0.07

Action Plan

RECOMMENDATIONS

01

TARGET

Shift marketing focus to **students and retired** clients. These demographics show significantly higher subscription rates (31% & 25%) compared to the general population.

HIGH CONVERSION SEGMENTS

03

RE-ENGAGE

Prioritize clients who subscribed in previous campaigns. This is the single most effective strategy with a **65.1% success rate**.

Prior Success	65.1%
New Contact	9.4%

02

TIMING

Execute major campaigns in **March, September, October, and December**. Avoid the high-volume, low-return month of May (6.4% success rate).

SEASONAL OPTIMIZATION

04

LIMIT

Cap outreach attempts at **2-3 calls** per prospect. Data shows a negative correlation (r = -0.07) between campaign frequency and success.

PREVENT FATIGUE

05

MONITOR

Use **Euribor rates** and employment numbers as market signals. High rates typically correlate with lower term deposit interest.

MACRO SIGNALS

Project Summary

CONCLUSION

Optimal Model

Random Forest outperformed all other algorithms, achieving robust predictive power by effectively handling non-linear relationships in the data.

Key Success Factors

Prediction accuracy relies heavily on **Duration, Poutcome (History), Month, and Customer Segment**. These four features drive the majority of the model's decision-making.

Handling Imbalance

The application of **SMOTE** (Synthetic Minority Oversampling) was critical in preventing bias towards the majority class (88.7% 'No'), ensuring the model can accurately identify potential subscribers.

Strategy Shift

Targeted campaigns focusing on high-probability segments (Students/Retired) outperform broad mass outreach strategies, offering significantly higher ROI.

82.06%

Balanced Accuracy

Random Forest

PERFORMANCE LEADER

"Smart targeting based on job type, timing, and past behavior can significantly improve term deposit subscription rates while reducing operational costs."

Academic Sources

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04 **Moro, S., Cortez, P., & Rita, P. (2014)**
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05 **UCI Machine Learning Repository**
Bank Marketing Dataset
archive.ics.uci.edu/ml/datasets/bank+marketing

Thank You!

Q & A